

# Sanket Sadashiv Hajare

📍 Pune, Maharashtra   📞 9322736663   ✉ sankethajare90@gmail.com   🔗 LinkedIn   🐙 GitHub

## ABOUT ME

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Aspiring **C Developer** with a strong foundation in **System Programming**, **UNIX Internals** and **Data Structures & Algorithms**. Hands-on experience in developing system utilities, concurrent networking applications and memory analysis tools using **C** and **POSIX interfaces**. Seeking an opportunity to design, develop and optimize software applications in a **Linux**-based development environment.

## EDUCATION

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| <b>Siddhant College of Engineering</b><br><i>Bachelor of Engineering - Computer Engineering; CGPA: 7.6</i> | Pune, Maharashtra<br>2022 - 2026 |
| <b>Shree Vinod Shivaji Bhate Jr. College</b><br><i>HSC (12th)</i>                                          | Sangli, Maharashtra<br>2021-2022 |

## PROJECTS

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| <b>Customised Virtual File System (CVFS)</b><br><i>Tech Stack: C, Operating System Concepts</i> <ul style="list-style-type: none"><li>Developed Linux-inspired virtual file system architecture replicating core operating system functionalities.</li><li>Implemented inode-based file management structure for efficient storage organization and retrieval.</li><li>Designed custom command interpreter supporting file creation, deletion, and access operations seamlessly.</li></ul> | 🐙 GitHub |
| <b>Kernel Interface Utility Suite</b><br><i>Tech Stack: C, Linux System Programming</i> <ul style="list-style-type: none"><li>Developed custom Linux command suite implementing 13+ utilities using native POSIX system calls in C.</li><li>Implemented core system calls including open, read, write, stat, fork, exec and chmod from scratch.</li><li>Integrated executables into system PATH with modular architecture and robust error handling mechanisms.</li></ul>                  | 🐙 GitHub |
| <b>Concurrent FTP Server</b><br><i>Tech Stack: C, Socket Programming, pthread</i> <ul style="list-style-type: none"><li>Developed multi-client concurrent FTP server using POSIX socket and pthread library APIs.</li><li>Implemented thread-per-client model enabling simultaneous file transfer and connection handling efficiently.</li><li>Designed TCP-based server architecture with robust concurrency control and synchronization mechanisms.</li></ul>                            | 🐙 GitHub |
| <b>Generalised Data Structures Library</b><br><i>Tech Stack: C++, Object-Oriented Programming</i> <ul style="list-style-type: none"><li>Developed reusable C++ library implementing linear and non-linear generic data structure templates.</li><li>Designed linked list, stack, queue and tree structures applying core object-oriented design principles.</li><li>Applied modular architecture enabling plug-and-play integration across multiple C++ application projects.</li></ul>    | 🐙 GitHub |

## TECHNICAL SKILLS

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**Programming Languages:** C, C++

**System Programming:** Process Management, Memory Management, File I/O, Signal Handling, System Calls

**Concurrency & IPC:** POSIX Threads (pthreads), System V IPC (Shared Memory, Semaphores, Message Queues)

**Network Programming:** TCP Socket Programming, Client-Server Architecture

**Operating System:** UNIX Internals, Virtual Memory, VFS, Buffer Cache, Scheduling, Boot Process

**CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, DBMS

**Development Practices:** Debugging (GDB), SDLC, Unit Testing, Code Review, Performance Optimization

**Version Control:** Git, GitHub

## CERTIFICATIONS & TRAINING

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| - <b>C / C++ Programming</b> – Marvellous Infosystems, Pune      | 2025 |
| - <b>Linux System Programming</b> – Marvellous Infosystems, Pune | 2025 |